SkyLedger

OPEN AGENT HACKATHON 2026 TRACK 1 · SOLVING FRAGMENTED INTELLIGENCE

# Satellites see the leaks.
# SkyLedger checks the reports.

An AI agent team that checks satellite-observed methane against what oil and gas facilities report to the US EPA, and who owns them. Every finding is ranged, traced, and challenged by a Skeptic before an analyst signs it.

- Zetaris
- Meterless Nightshift
- Meterless World Model
- NVIDIA Nemotron
- Cursor

github.com/skyledger-org/skyledger

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# Three sources that were never designed to meet.

SKY · CARBON MAPPER

## kg / hour

### A plume is a point and a rate

Seen from aircraft and satellites (Tanager, EMIT, AVIRIS-NG, GAO) on one morning. For example **`207 kg/h`** on 2026-09-05.

GROUND · US EPA GHGRP

## t $CO_2e$ / yr

### A report is a facility and a year

Self-reported once a year, sometimes for a whole basin. Tiger Plant reported **`16 t CH₄`** for RY2023.

LEDGER · OWNERSHIP

## free text

### An owner is a string

`"TARGA RESOURCES (73%); PIONEER NATURAL RESOURCES USA INC (27%)"`

Spelled many ways, no IDs.

**18** of 60

Permian plumes where matching to the nearest facility blames a plant **more than 1 km from where the plume starts**. No shared keys, units or clock, so nobody checks one against the other.

# For the people who must judge whether a reported number is plausible.

## ESG and credit analysts

Signing off on issuers' emissions disclosures, and pricing transition risk.

## Regulators and state agencies

Deciding which facilities to inspect first, with evidence that survives challenge.

## Journalists and NGOs

Who need traceable evidence, not anecdotes, before naming a company.

WHY NOW

## 10 Nov 2026

California **SB 253**: first Scope 1 and 2 reports are due, with no assurance requirement yet.

## RY2025

EPA has **delayed GHGRP reporting** and proposed ending most of the program. Independent checks matter more.

## Public & frequent

Satellite methane data is now open (Carbon Mapper, Tanager).

# Every plume becomes a traced, ranged, signed-off dossier.

### 1 · Find

Federated geo-join of plumes to reporting facilities, queried in place through Zetaris.

### 2 · Attribute

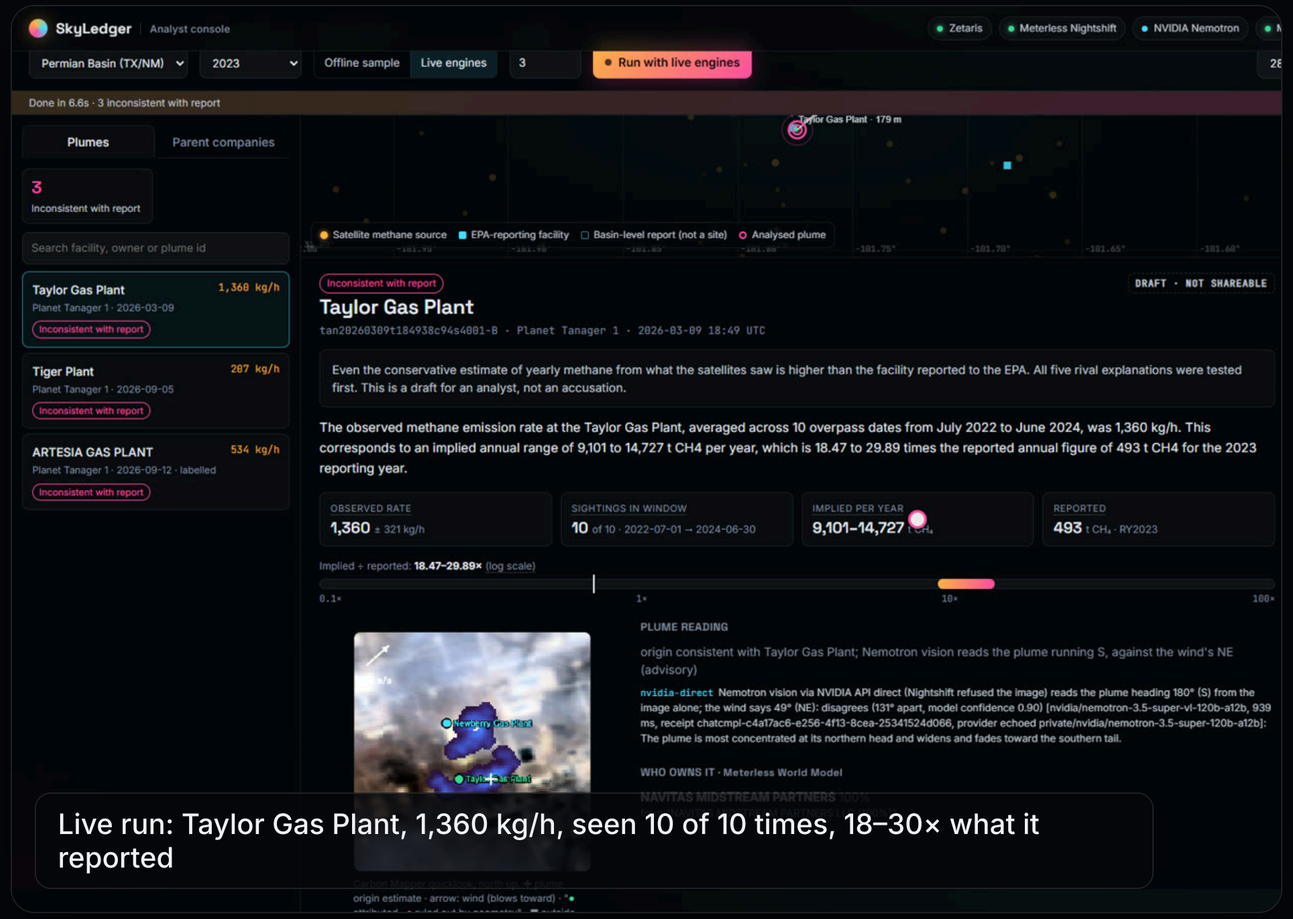
Wind and plume geometry pick the facility at the plume's head. Nemotron reads the image as a check.

### 3 · Reconcile

kg/h becomes tonnes per year, always a range, inside the reporting window. Owners resolve to parents.

### 4 · Challenge and sign

A Skeptic rules out five innocent explanations first. Nothing is shareable until an analyst signs.




Live run: Taylor Gas Plant, 1,360 kg/h, seen 10 of 10 times, 18–30× what it reported

# Eight agents, one shared state, and a Skeptic with veto power.

**SourceMapper**
what's reachable

→

**GeoLinker**
Zetaris geo-join

→

**PlumeReader**
wind + Nemotron

→

**Ownership**
World Model

→

**Reconciler**
units, window

**Skeptic**
5 rival tests · blocks or sends work back

→

**Reporter**
Nemotron via Nightshift + guard

→

**SignOff**
signed execution contract

## Not a linear chain

The Skeptic sends work back: widen the search (GeoLinker), look again (PlumeReader), or stop annualising (Reconciler). Loops are bounded.

## Numbers come from code

Distances, unit conversion and ratios come from SQL and code. Models read images and word findings; a guard rejects any number the code didn't compute.

## Traceable

Every step goes to a hash-chained JSONL trace that streams live to the console. Editing any line breaks the chain.

# Before a word is written, five rival explanations must fall.

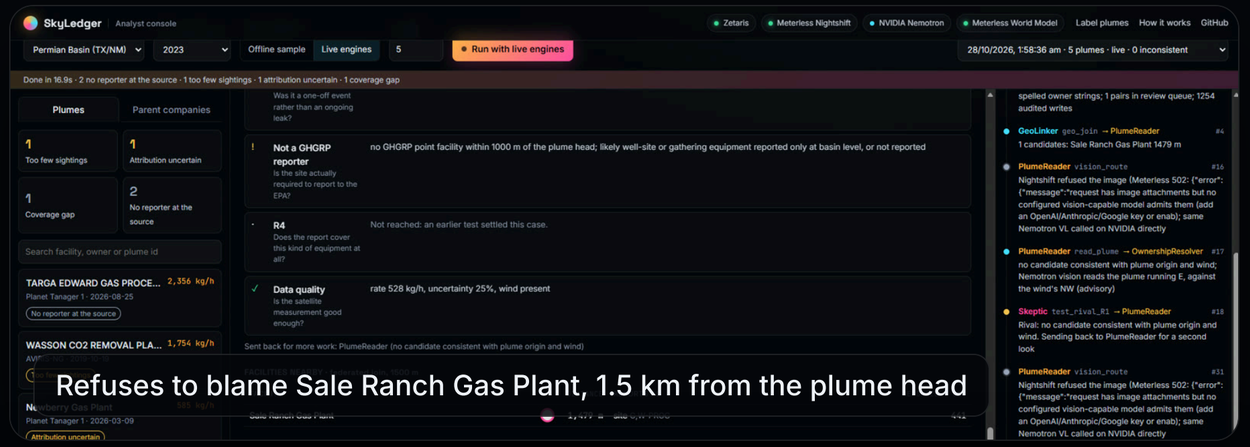
**R1** **Wrong facility**
Another site fits the plume head and wind, or the nearest reporter is over 1 km away.

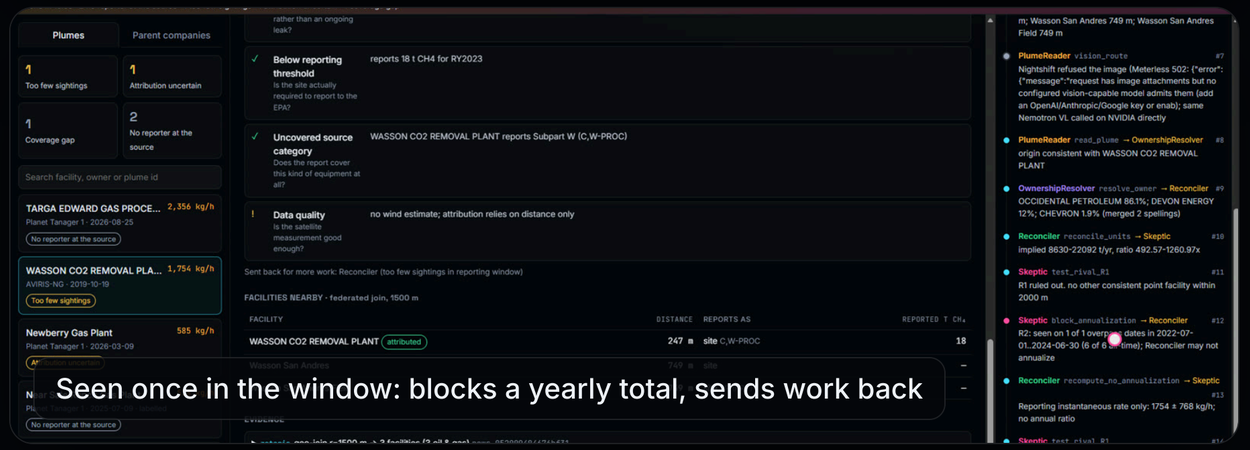
**R2** **One-off release**
Fewer than 2 sightings in the reporting window blocks any yearly total.

**R3** **Not required to report**
No subparts, no $CH_4$ filed, or no reporter at the plume head.

**R4** **Uncovered equipment**
Only basin-level reports nearby, or no Subpart W coverage.

**R5** **Bad measurement**
No rate, uncertainty above the limit, or no wind.



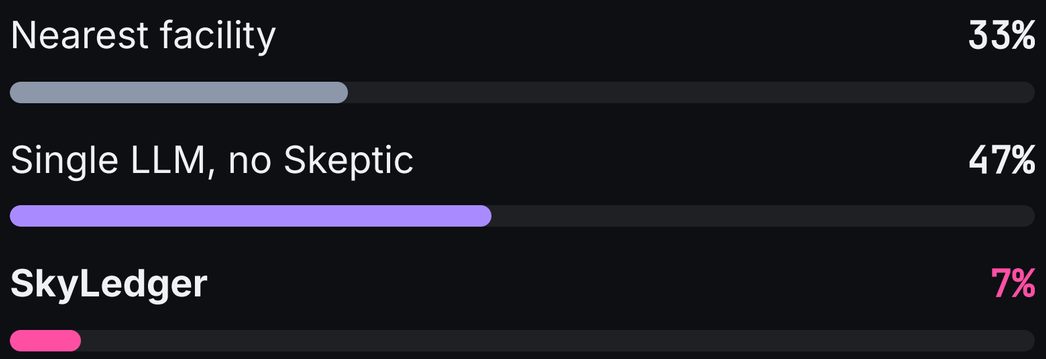
# Fewer flags. Better ones.

DISCREPANCIES FLAGGED, 60 PERMIAN PLUMES

## 34 → 8

A naive nearest-facility, all-time-persistence approach flags 34. Eight survive the Skeptic, each with a range, the exact SQL and evidence.

WRONG FACILITY BLAMED, 15 HAND-LABELLED PLUMES




A small sanity check labelled by the team, not expert ground truth. Details in eval/results.md.

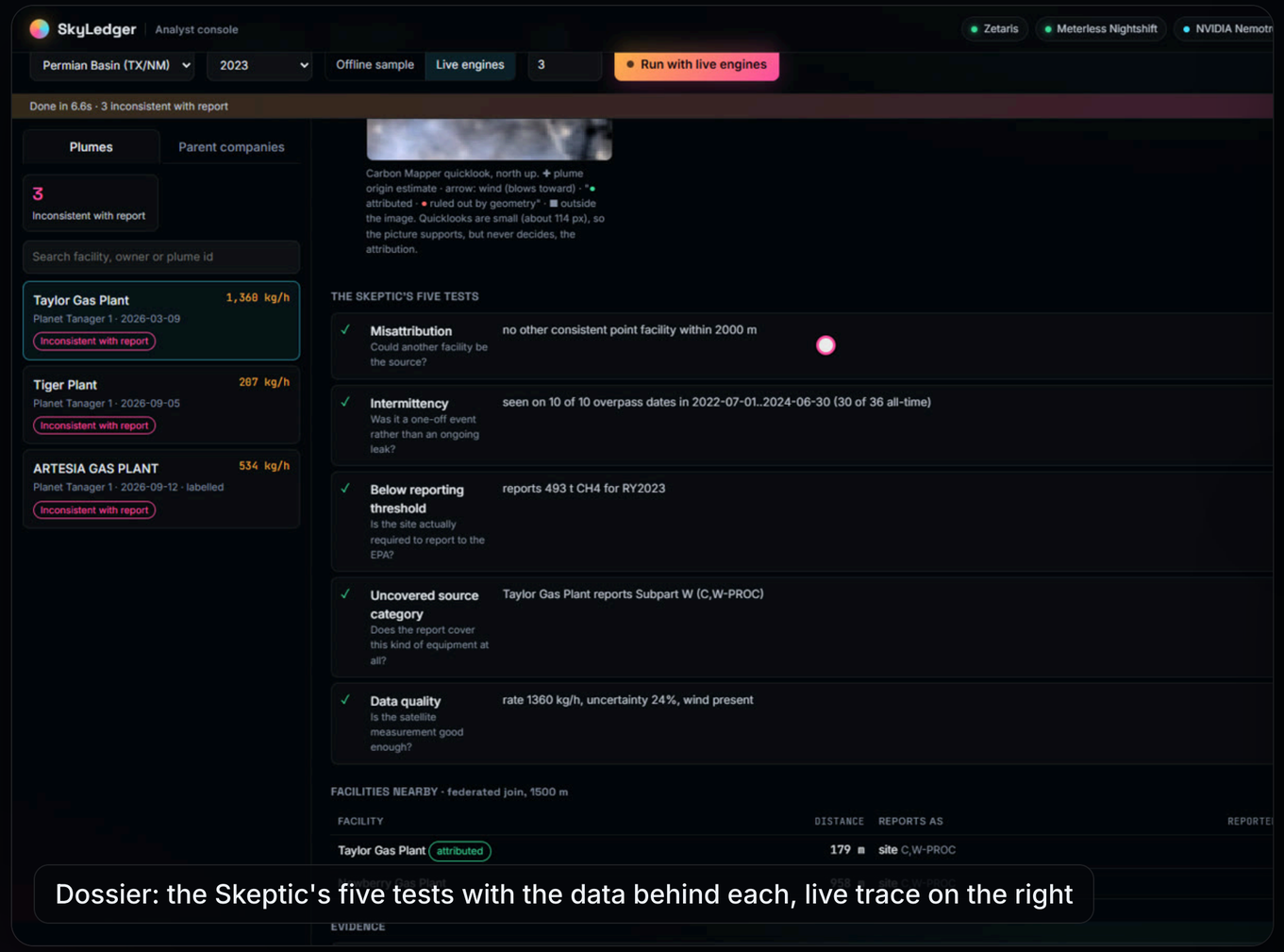
**3,413**
satellite sources in the scan area

**31**
plumes with no reporting facility at the head

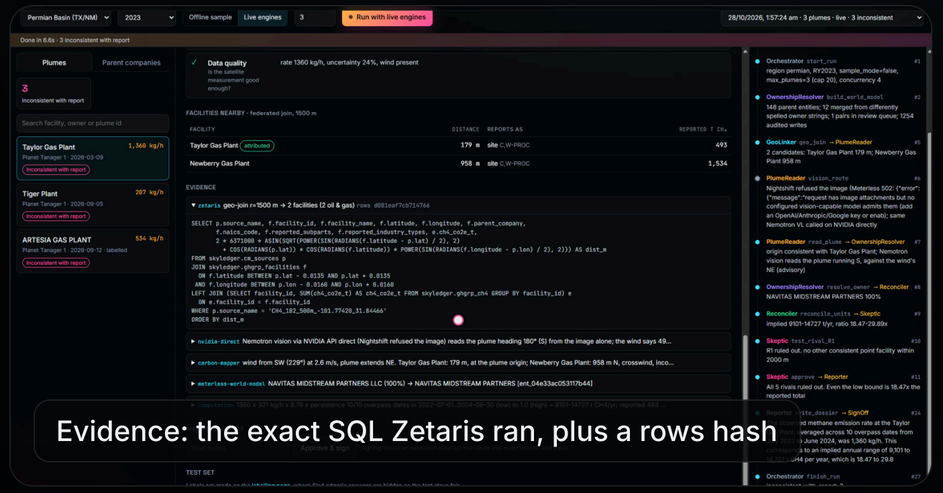
**10**
too few sightings to estimate a year

**0**
false discrepancy flags on the labelled set

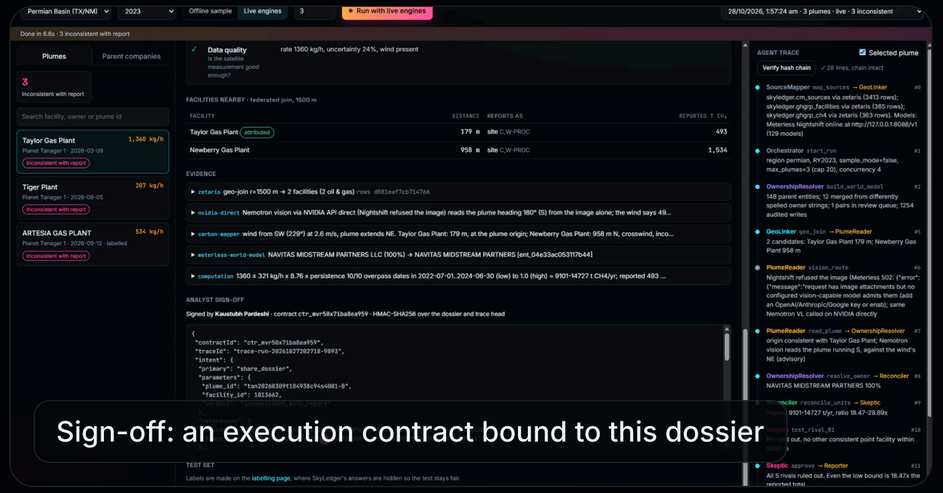
# An analyst console, not a chatbot.




Dossier: the Skeptic's five tests with the data behind each, live trace on the right



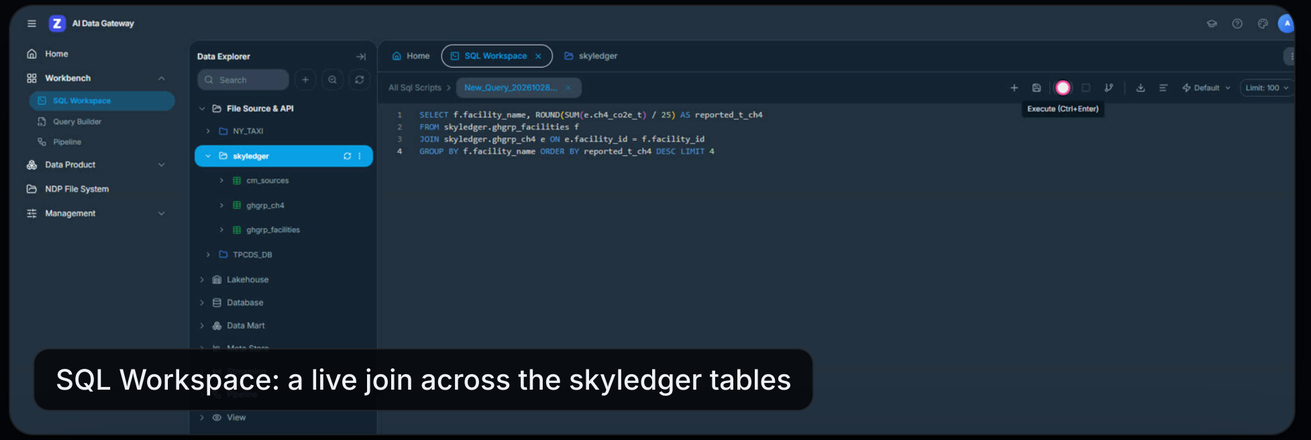

Evidence: the exact SQL Zetaris ran, plus a rows hash



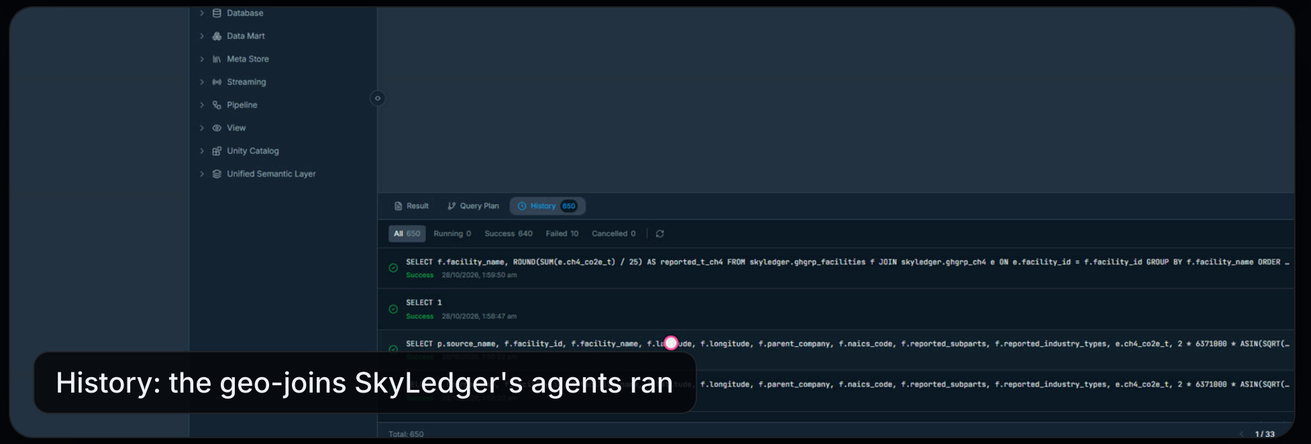

Sign-off: an execution contract bound to this dossier

# Zetaris is the data layer: joined in place, never copied.

- **Three public sources** registered as Zetaris Lightning tables: Carbon Mapper sources, GHGRP facilities, GHGRP $CH_4$ by subpart.
- **One federated geo-join per plume** through the REST SQL API: bounding box plus haversine distance, where the data lives.
- **SourceMapper** discovers the reachable tables and row counts at the start of every run.
- **Evidence:** the exact SQL and a rows hash in every dossier, so an analyst can rerun it.
- **Verified:** Zetaris equals the offline cache for **60 / 60** plumes, and the agents' queries appear in Zetaris's own history.

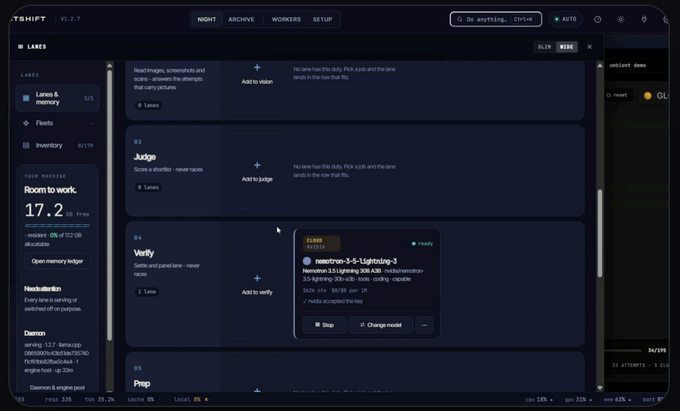



SQL Workspace: a live join across the skyledger tables



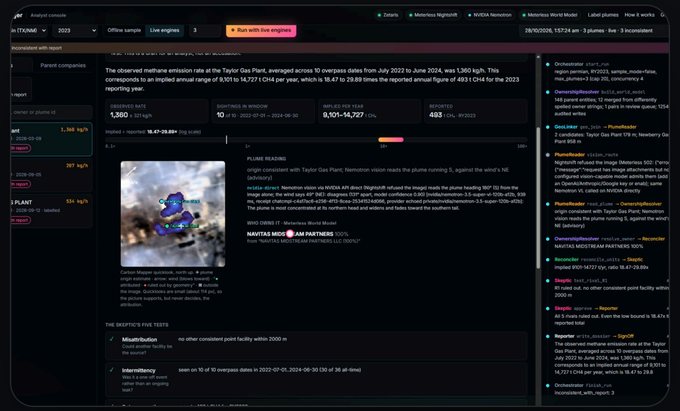

History: the geo-joins SkyLedger's agents ran

# Meterless runs the models and the ownership graph. Nemotron reads and writes.



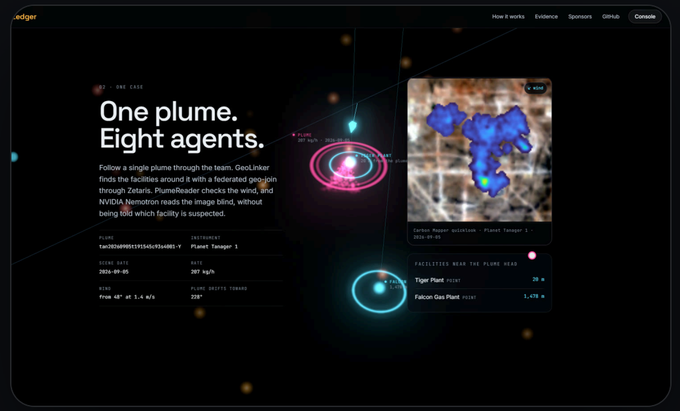

## Meterless Nightshift

The primary AI provider, built to Meterless's provider contract (headers, discovery, token handling, Auto Detect). The NVIDIA key lives inside Nightshift, not in our app. Contract checks pass.




## Meterless World Model

The facility → owner → parent graph, with provenance on every write, merged spellings and a review queue. Meterless's conformance suite: **8 / 8**. Sign-off uses Scout Intent's contract format.




## NVIDIA Nemotron

**3.5 Lightning** via Nightshift writes each dossier under a number guard. **3.5 Super VL** reads the plume image blind: 38% agreement with the wind (chance ≈ 33%), so it's advisory. VL fallback to NVIDIA's API is disclosed and logged.

# Small, reproducible, and honest about what's live.

STACK

- **TypeScript on Node 22**, run with tsx. No build step, no GPU.
- **HTTP API:** POST /run, live server-sent-event streaming, signed sign-off.
- **Front end:** vanilla JS, three.js, GSAP and Lenis, served locally.
- **Config, not constants:** every threshold in config/skyledger.json, regions in config/regions/.
- **Sample mode and Docker:** runs from a clean clone with no keys and no network.

MEASURED, NOT CLAIMED

| | |
|---|---|
| 60/60 | Zetaris geo-join equals the offline cache |
| 8/8 | Meterless World Model conformance suite |
| 38% | Nemotron vision vs the wind (chance ≈ 33%), so advisory |
| 7% | wrong facility blamed on hand labels (nearest facility: 33%) |
| ✓ | clean-clone Docker run, no .env, no network |

11 FUTURE ENHANCEMENTS

# From one basin to a monitoring service.

## Coverage

Every US basin and reporting year, scheduled refresh from Carbon Mapper and EPA, SEC CIK links for parent companies.

## Accuracy

A larger expert-labelled test set, vision on larger scenes as a data-quality check, and learning attribution rules from analyst decisions.

## Product

A hosted service with logins and audit trails, alerts when a facility's gap persists, and signed dossiers exported for disclosure reviews.

**SkyLedger**

Check what satellites saw against what was reported.

`github.com/skyledger-org/skyledger`
`runs offline: npm ci && npm start`



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